

CELL BIOLOGY

ESCRT III repairs nuclear envelope ruptures during cell migration to limit DNA damage and cell death

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In eukaryotic cells, the nuclear envelope separates the genomic DNA from the cytoplasmic space and regulates protein trafficking between the two compartments. This barrier is only transiently dissolved during mitosis. Here, we found that it also opened at high frequency in migrating mammalian cells during interphase, which allowed nuclear proteins to leak out and cytoplasmic proteins to leak in. This transient opening was caused by nuclear deformation and was rapidly repaired in an ESCRT (endosomal sorting complexes required for transport)–dependent manner. DNA double-strand breaks coincided with nuclear envelope opening events. As a consequence, survival of cells migrating through confining environments depended on efficient nuclear envelope and DNA repair machineries. Nuclear envelope opening in migrating leukocytes could have potentially important consequences for normal and pathological immune responses.

The nuclear envelope (NE) functions as a barrier to separate the chromatin from the cytoplasm and is considered to remain intact during interphase. Only under pathological circumstances is the NE thought to open in nonmitotic cells. It can bud during viral infection (1) or be completely breached in laminopathies, pathologies associated with mutations in genes coding for nuclear lamina proteins, especially in LMN A/C (2, 3). Many cancer cells express lower levels of LMN A/C, which correlates with a higher degree of metastatic potential (4). This is because nuclei lacking LMN A/C can be more easily deformed (5), which allows them to migrate through narrower pores and to invade tissues (6). Complete removal of LMN A/C leads to an increase in cell death during migration through pores and, eventually, reduces the extent of metastasis (4). The cause of this cell death remains unknown. Similarly to cancer cells, several types of immune cells express lower levels of LMN A/C. These cells also have the capacity to migrate through dense tissues (7). Their response to pathogens is tightly associated with their migratory capacities. Do specific survival mechanisms exist to allow highly migratory cells to survive their journey through tissues, despite a large degree of nuclear deformation?

We compared bone marrow–derived mouse dendritic cells (mDCs), migrating between two surfaces spaced 5 μm apart either without (Fig. 1A and fig. S1A) or with collagen filling (Fig. 1B), as well as cells migrating through mice ear explants (Fig. 1C and fig. S1B). In collagen or ear explants, nuclei were more irregularly shaped (Fig. 1D), and the minimum diameter of the nucleus was reduced (Fig. 1E). This reflected pinched or dumbbell-shaped nuclei (arrows, Fig. 1C), as has been observed previously in migrating cancer and immune cells (8, 9). Thus, DCs deform their nucleus during migration through physiological environments.

We recorded mDCs expressing green fluorescent protein (GFP) linked to a nuclear localization sequence (NLS-GFP). We followed cells migrating through a collagen gel (fig. S1C), or in an ear explant (fig. S1, D and E, and movie S1). Cells undergoing strong nuclear deformations displayed a decrease in the GFP nuclear signal and an increase in the cytoplasmic signal (Fig. 1F). This suggests that the nucleo-cytoplasmic barrier was transiently abrogated and then restored, which likely corresponds to an opening of the NE.

Collagen gels and ear explants offer poor control over the degree of nuclear deformation. We thus used a migration assay consisting of microchannels with constrictions of various sizes (fig. S2, A and B) (10), which matched the range of sizes observed in vivo for nuclear deformation (Fig. 1E) (11). To extend our findings to human cells, we investigated monocyte-derived human DCs (hDCs) (Fig. 2A and movie S2), as well as cultured cancer cells (HeLa) (Fig. 2B, fig. S2C, and movie S2) and normal cultured immortalized cells (RPE-1) (fig. S2D and movie S2). The NLS-GFP nuclear signal strongly de-

creased and the cytoplasmic signal increased while the nucleus was crossing the constriction (fig. S3, A and B). Cytoplasmic signal remained high as long as the nucleus was engaged within the constriction, independently of cell speed.

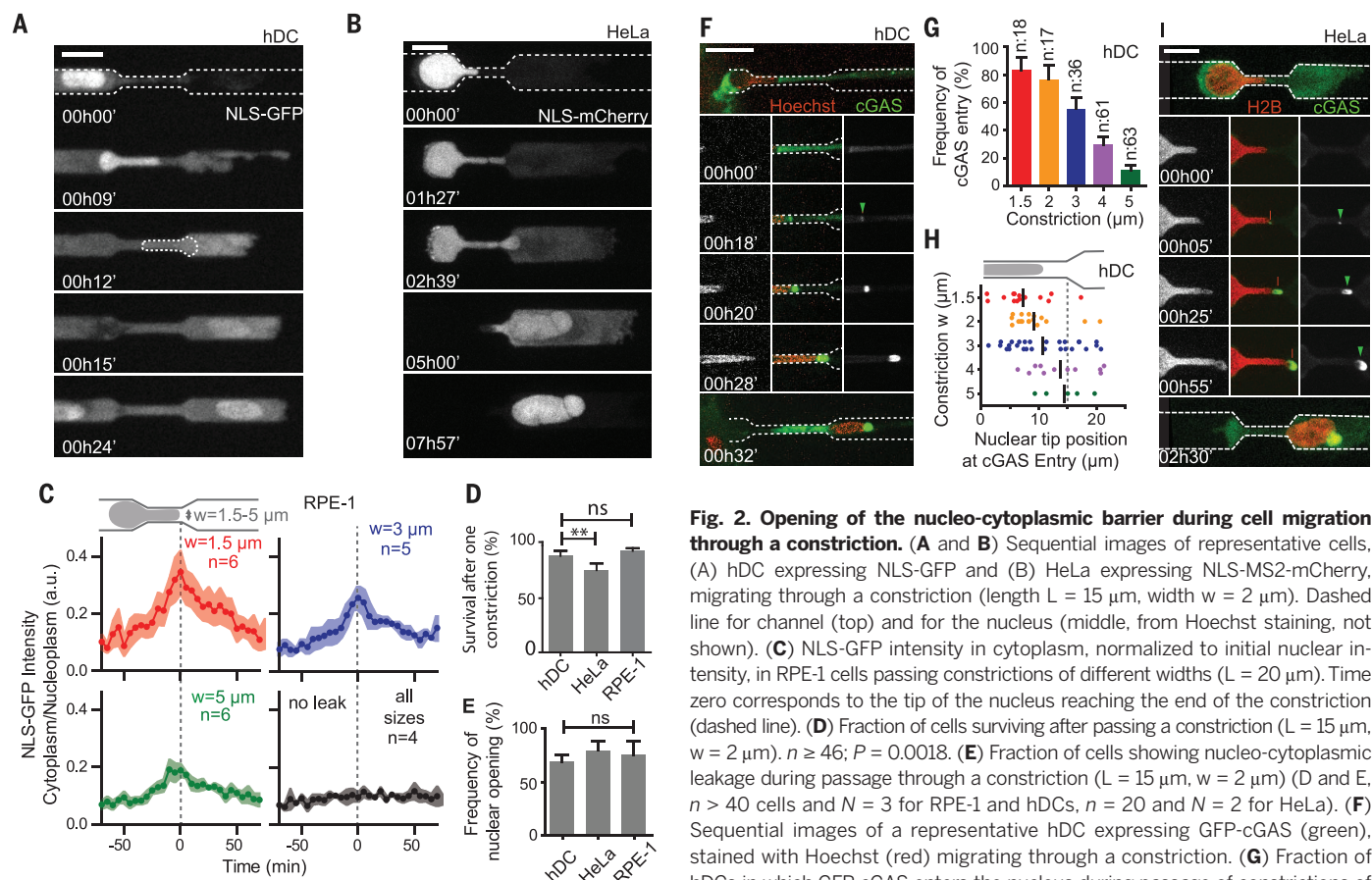
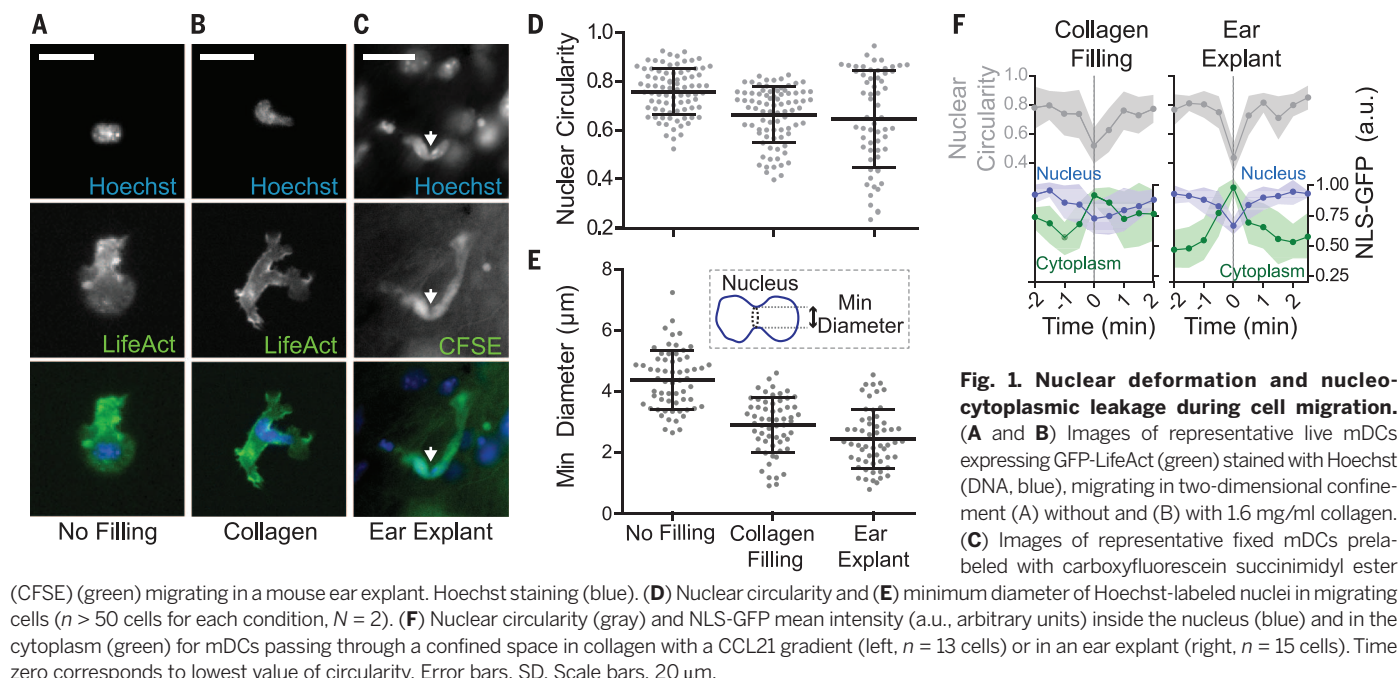
The slow passage of HeLa cells allowed us to observe several individual leakage events (bursts in fig. S3, B to E), which corresponded to the formation and rapid disappearance of bleblike structures at the tip of the passing nucleus (fig. S3, B to D, and movie S2). This suggests a mechanism by which deformation of the nucleus generates increased internal pressure and the formation of NE blebs that eventually rupture, followed by a resealing process, until the next bleb forms and ruptures. This is reminiscent of compressed nonmigrating cells (12). Consistently, the amount of NLS-GFP cytoplasmic leakage increased when the constriction size was narrower (Fig. 2C). Thus, migrating cells, when deforming their nucleus, display a high survival rate (Fig. 2D and fig. S3F) despite frequent opening of their NEs (Fig. 2E).

To assess the precise timing and location of NE opening, we used cells expressing a cytoplasmic DNA binding probe fused to GFP [cyclic guanosine monophosphate–adenosine monophosphate (GMP-AMP) synthase, cGAS]. Upon compression of hDCs, nuclear blebs were induced (12) (fig. S4A and movie S3). GFP-cGAS localized to chromatin at the exact location where the nuclear bleb had formed and ruptured (fig. S4A). Using this probe, we confirmed that hDCs opened their NEs (Fig. 2F and movie S4) at a higher frequency (Fig. 2G and fig. S4B) and earlier in the constriction (Fig. 2H and fig. S4C) for smaller constrictions, and that openings were mostly localized at the front tip of the nucleus. This was also observed during bleb bursting in HeLa cells (Fig. 2I and fig. S4, D to G). Staining of NE components showed that the nuclear lamina also ruptured (fig. S5, A and B) and that nuclear pores were excluded from the rupture region at the tip of the nucleus (fig. S5, B and C). Recording GFP-tagged Lap2 β (Lap2 β -GFP), an inner NE protein bound to the nuclear lamina, confirmed that the NE formed blebs devoid of lamina, that blebs eventually ruptured, and that they could contain chromatin (fig. S5, D to F, and movie S5). These observations are consistent with a simple physical model of deformation-induced blebbing followed by NE opening and resealing (fig. S6).

Plasma membrane repair and postmitotic NE resealing both require the ESCRT III complex (13–15). We therefore used cells coexpressing GFP-tagged CHMP4B (CHMP4B-GFP) (16), an ESCRT III complex subunit, and RFP-cGAS. Nearly all nuclear blebs that burst subsequently recruited CHMP4B-GFP (95%; $n = 62$) (fig. S7A and movie S6). We could also induce recruitment of CHMP4B-GFP using laser ablation aimed at the nuclear edge (fig. S7, B to D; fig. S8; and movie S7), associated with NLS-mCherry leakage into the cytoplasm (fig. S9). In cells migrating through constrictions, CHMP4B-GFP was transiently recruited to sites of

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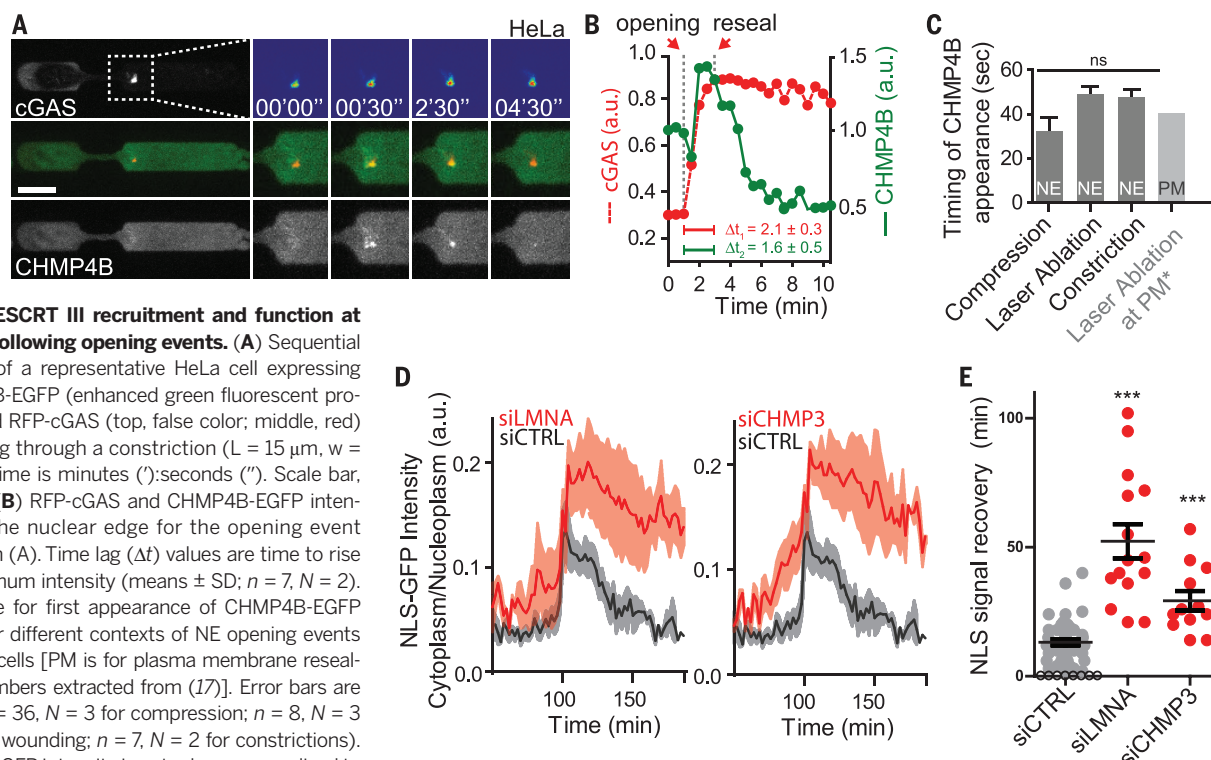


Fig. 3. ESCRT III recruitment and function at the NE following opening events. (A) Sequential images of a representative HeLa cell expressing CHMP4B-EGFP (enhanced green fluorescent protein) and RFP-cGAS (top, false color; middle, red) migrating through a constriction ($L = 15 \mu\text{m}$, $w = 2 \mu\text{m}$). Time is minutes (:):seconds ("). Scale bar, $20 \mu\text{m}$. (B) RFP-cGAS and CHMP4B-EGFP intensity at the nuclear edge for the opening event shown in (A). Time lag (Δt) values are time to rise to maximum intensity (means $\pm$ SD; $n = 7$, $N = 2$). (C) Time for first appearance of CHMP4B-EGFP at NE for different contexts of NE opening events in HeLa cells [PM is for plasma membrane resealing, *numbers extracted from (17)]. Error bars are SEM ($n = 36$, $N = 3$ for compression; $n = 8$, $N = 3$ for laser wounding; $n = 7$, $N = 2$ for constrictions). (D) NLS-GFP intensity in cytoplasm, normalized to initial nuclear intensity, in RPE-1 cells treated with control small interfering RNA (siRNA); siCTRL, black curves; and with LMNA (left) and CHMP3 (right) siRNAs (red curves). $n \geq 6$ cells for each curve ($N = 2$). Error bars are SEM. (E) Quantification of the time needed for full recovery of NLS-GFP in the nucleus. Time zero is nuclear exit from the constriction (open circles are for recovery before exit; $n = 46$, 15 , and 12 ; $N = 2$; error bars, SEM; *** $P < 0.0001$).

RFP-cGAS entry, at the nucleus tip (Fig. 3A and movie S8). CHMP4B-GFP localized to the site of NE opening just after the opening occurred and decreased after resealing (Fig. 3B and fig. S10, A and B). Thus, NE opening induced with these three different methods (compression, laser, and confined migration), recruited CHMP4B-GFP with kinetics similar to those observed after plasma membrane wounding (Fig. 3C) (13, 17). This suggests that the resealing of the NE after opening caused by nuclear deformation in migrating cells might also require the ESCRT III complex machinery.

To test the function of ESCRT III in NE resealing, we depleted (knocked down) CHMP3 (14), which delayed recruitment of CHMP4B-GFP after laser wounding (fig. S10C). We also knocked down LMN A/C, whose depletion, as expected (18, 19), caused random bursts of NLS-GFP into the cytoplasm even in the absence of nuclear constriction (fig. S10D and movie S9). CHMP3-depleted cells did not show any leakage of NLS-GFP in the cytoplasm outside constrictions, which indicated that CHMP3 depletion did not make the NE more susceptible to spontaneous opening. However, when CHMP3-depleted cells passed through constrictions, they showed increased GFP signal in the cytoplasm, as did LMN A/C-depleted cells, and the cytoplasmic signal remained for prolonged periods of time after the cells passed the constriction (Fig. 3, D and E; fig. S10, E to G; fig. S11, A and B; and movie S9). Thus the ESCRT

III complex is essential to reseal the NE in migrating cells.

We hypothesized that efficient resealing was responsible for the high survival rate of cells passing through constrictions. As expected (4), we observed increased death in LMN A/C-depleted cells (Fig. 4A), even inside straight channels, but more so when cells were passing constrictions. CHMP3 depletion did not induce any increase in cell death (Fig. 4A); thus, prolonged nuclear opening alone was not enough to cause cell death. LMN A/C-depleted cells exhibit defects in DNA damage response (20). We thus imaged RPE-1 cells expressing 53BP1-GFP, a protein recruited to DNA double-strand breaks (21). These cells showed a transient increase in the number and intensity of 53BP1-GFP foci during passage of the nucleus through a constriction (Fig. 4, B to D, and movie S10), with kinetics similar to NLS-GFP leakage into the cytosol. This suggested that DNA double-strand breaks might result from NE opening, although we cannot fully exclude that mechanical constraints exerted on the DNA during nuclear deformation also play a role. We imaged cells expressing both 53BP1-GFP and RFP-cGAS and found that the formation of 53BP1-GFP foci always followed the recruitment of RFP-cGAS into the nucleus (Fig. 4, B and E). Furthermore, only cells that exhibited entry of RFP-cGAS into the nucleus also showed an increase in 53BP1-GFP foci (Fig. 4F), which were dispersed throughout the nucleus (Fig. 4G). This suggests that

diffusing cytoplasmic factors might enter during NE opening, and in turn induce DNA damage. 53BP1-GFP foci disappeared a few tens of minutes after cells exited the constriction, which implies efficient DNA repair. We thus inhibited DNA repair using an ataxia telangiectasia mutated (ATM) inhibitor (ATMi) (fig. S11C) (22). Upon ATMi treatment, the level of cell death was increased only in LMN A/C- and in CHMP3-depleted cells migrating through constrictions (Fig. 4A). Together these experiments show that DNA damage caused by prolonged NE opening can lead to cell death provided that DNA repair is also affected.

Here, we have shown that nuclear deformation during cell migration leads to transient opening of the NE and that the ESCRT III complex, similarly to what happens at mitotic exit (14, 15), is required for fast resealing of the nucleo-cytoplasmic barrier (fig. S12). This transient opening leads to nucleo-cytoplasmic mixing, which potentially causes DNA damage owing to entry of cytoplasmic factors (fig. S13) (23–26). We propose that this phenomenon might be particularly relevant for immune and cancer cells (27). We anticipate that in various developmental, immunological or pathological contexts, nuclear deformation-associated NE rupture could lead to a large range of cellular responses, as well as to genomic instability, cell aging, or cell death (28–30). Such responses could be important physiologically or pathologically.

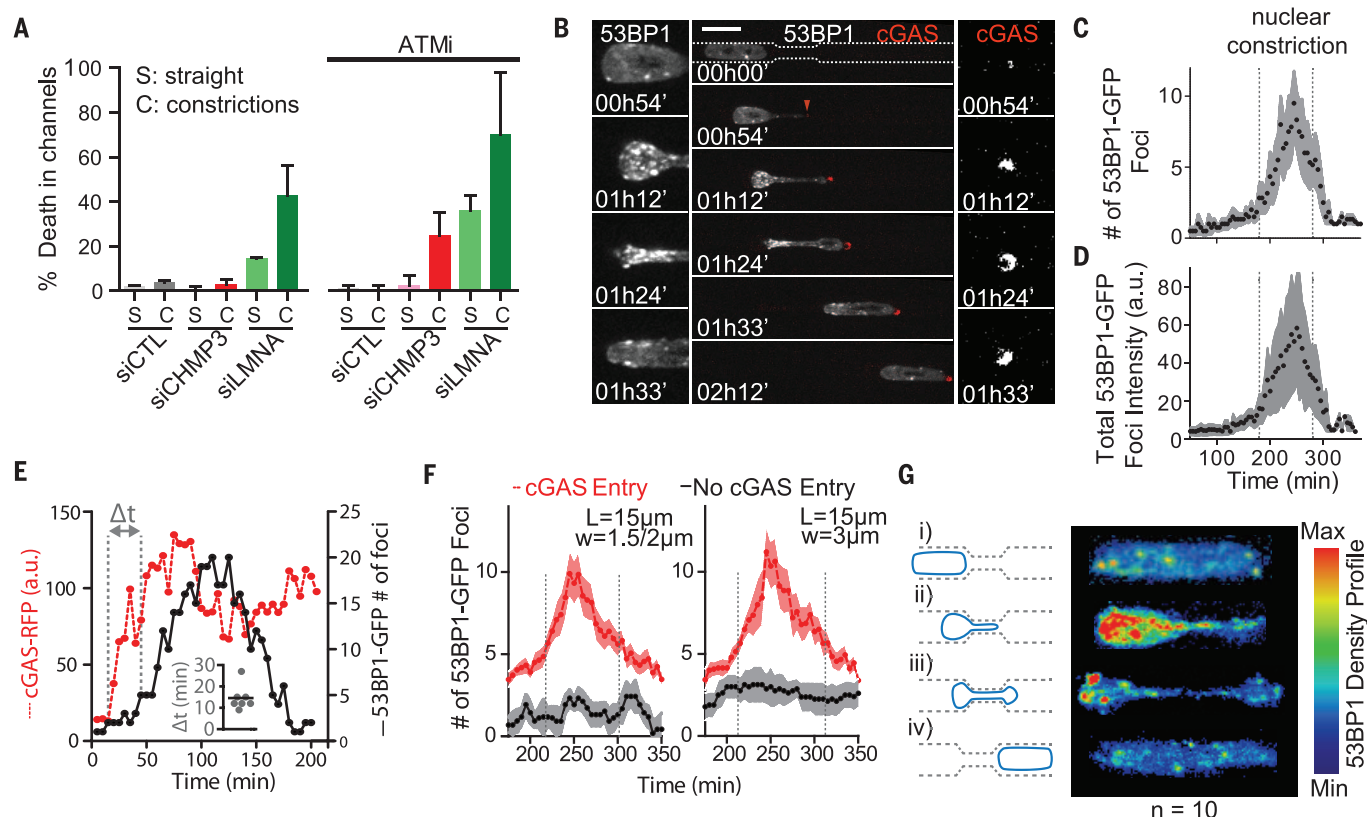


Fig. 4. DNA damage associated with nuclear deformation and NE opening.

(A) Fraction of RPE-1 cells dying after passing one constriction. siCTL, siRNA control. “Straight” is for cells moving across the same distance in channels without constrictions (L = 20 μm, w = 3 μm; left to right, n = 300, 300, 240, 240, 90, 90, 300, 300, 120, 60, 60, 60; N = 3 for each condition). (B) Sequential images of a representative RPE-1 cell expressing 53BP1-GFP (gray levels) and RFP-cGAS (red) migrating through a constriction (L = 15 μm, w = 2 μm). Red arrow shows first RFP-cGAS entry at nuclear tip. Time is hours (h):minutes ('); scale bar, 20 μm. Number (C) and total intensity (D) of 53BP1-GFP foci in the nucleus while passing a constriction (L = 15 μm, w = 1.5 μm; n = 7 cells, N = 3). (E) RFP-cGAS intensity (red

curve) inside the nucleus and number of 53BP1-GFP foci (black curve) for a representative cell. (Inset) Time lag (Δt) between the entry of RFP-cGAS and the increase in number of 53BP1-GFP foci (n = 7, N = 2). (F) Number of 53BP1-GFP foci in the nucleus, for cells which showed entry of RFP-cGAS in the nucleus (red curve) compared with cells that did not show entry of RFP-cGAS (black curve), for the same size of constrictions (L = 15 μm, left: w = 1.5 μm and 2 μm, n = 25 and 5 cells; right: w = 3 μm, n = 13 and 9; N = 3). (G) 53BP1-GFP density profile in nuclei at different stages of passing constrictions: (i) before entering, (ii) nuclear tip reaching the end of the constriction, (iii) nucleus halfway through the constriction, and (iv) after exiting (N = 2). Constrictions are (L = 15 μm, w = 2 μm). Error bars, SEM.

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SUPPLEMENTARY MATERIALS

www.sciencemag.org/content/352/6283/359/suppl/DC1
Materials and Methods
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Movie Captions S1 to S10
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Movies S1 to S10

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